

Linear algebra – 1

Determinants and invertibility

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Theoretical minimum.

Definition. The adjugate $\text{adj } A$ has entries $(\text{adj } A)_{ij} = (-1)^{i+j} M_{ji}$, where M_{ji} is the corresponding minor; it satisfies $A \text{adj } A = \text{adj } A \cdot A = (\det A)E$, and $\text{rank adj } A$ equals n , 1 or 0 according as $\text{rank } A$ equals n , $n - 1$, or less.

Theorem 1 (Jacobi's formula). $\frac{d}{dt} \det A(t) = \text{tr}(\text{adj } A(t) \cdot A'(t))$; in particular $\det(E + tB) = 1 + t \text{tr } B + O(t^2)$ and $\det e^A = e^{\text{tr } A}$. The same expansion gives $\text{adj}(\text{adj } A) = (\det A)^{n-2} A$ for $n \geq 2$.

Theorem 2 (rank-one update). For invertible A ,

$$\det(A + uv^T) = \det A \cdot (1 + v^T A^{-1}u),$$

and more generally $\det(A + UV^T) = \det A \cdot \det(E_k + V^T A^{-1}U)$ for $n \times k$ blocks. A matrix of the form “diagonal plus constant” or “ $\cos((j+k)\theta)$ ” is a small-rank perturbation of something trivial, and this formula is the shortest way to its determinant.

Theorem 3 (invertibility over a ring). An integer matrix has an integer inverse if and only if $\det A = \pm 1$; over $\mathbb{Z}_m$ invertibility means $\det A$ invertible modulo m . Since $\det$ is a polynomial with integer coefficients in the entries, every determinant identity valid over $\mathbb{Z}$ is valid over any commutative ring, and reduction modulo a prime is a legitimate way to prove that a determinant is non-zero.

Fact 1 (three ways to prove singularity). $\det A = 0$ if and only if the columns are dependent, if and only if 0 is an eigenvalue, if and only if $Ax = 0$ has a non-trivial solution. The first is combinatorial, the second spectral

(Linear algebra – 3), the third analytic; almost every olympiad problem about invertibility is solved by choosing the right one of the three.

1. Prove Jacobi's formula, deduce $\det e^A = e^{\text{tr } A}$, and prove the identity $\text{adj}(\text{adj } A) = (\det A)^{n-2} A$ for $n \geq 2$.
2. (Putnam 2014) Let A be the $n \times n$ matrix whose entry in the i -th row and j -th column is $\frac{1}{\min(i,j)}$ for $1 \leq i, j \leq n$. Compute $\det A$.
3. (Putnam 1992) Let D_n denote the value of the $(n-1) \times (n-1)$ determinant of the matrix whose diagonal entries are $3, 4, 5, \dots, n+1$ and all of whose off-diagonal entries equal 1. Is the set $\{D_n/n!\}_{n \geq 2}$ bounded?
4. (Putnam 1999) For an integer $n \geq 3$, let $\theta = \frac{2\pi}{n}$. Evaluate the determinant of the $n \times n$ matrix $E + A$, where $A = (a_{jk})$ has entries $a_{jk} = \cos((j+k)\theta)$ for all j, k .
5. (Romania national olympiad 2006) Let $A \in M_n(\mathbb{C})$ and suppose that $(\text{adj } A)^m = O$ for some positive integer m . Prove that $(\text{adj } A)^2 = O$.
6. (Putnam 1991) Let A and B be different $n \times n$ matrices with real entries. If $A^3 = B^3$ and $A^2B = B^2A$, can $A^2 + B^2$ be invertible?
7. (Putnam 1994) Let A and B be 2×2 matrices with integer entries such that $A, A+B, A+2B, A+3B, A+4B$ are all invertible matrices whose inverses have integer entries. Show that $A+5B$ is invertible and that its inverse has integer entries.
8. (MexMat, 2009) Prove that every linear map from the space of complex $n \times n$ matrices to itself which preserves the determinant is invertible.